Protocol design elements and USDM

Data or words?

What USDM holds for objectives, endpoints, eligibility criteria and the schedule

Data | Data in a string | Words

Kerstin Forsberg · the published USDM v4 model and its CDISC example files, represented as a knowledge graph

A first look

The model is mostly structure. In a real study, the words sit in a few places.

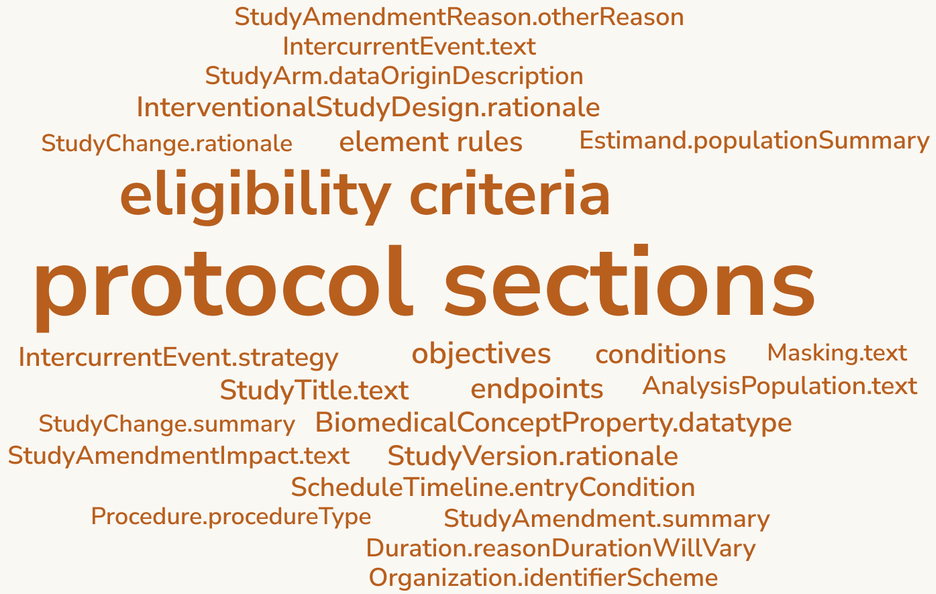
The USDM v4 model

Attribute names, sized by how many classes have them



A real study: where the words are

Text in the CDISC Pilot example, sized by characters



In the CDISC Pilot, 238 text values hold 82% of the characters, against 4,112 values that are data (housekeeping and naming not counted). The model cloud leaves out housekeeping and containment attributes.

Data | Data in a string | Words | Naming

The question

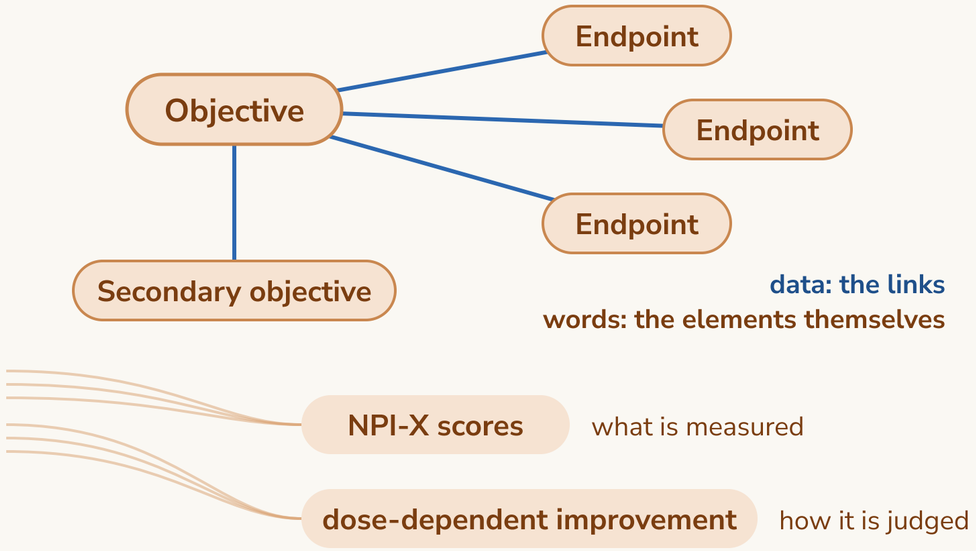
For each design element: which parts are data, and which are still words?

AN OBJECTIVE IN USDM · CDISC PILOT STUDY EXAMPLE

Level: secondary objective | Its 3 endpoints, linked

> “To assess the dose-dependent improvement in behavior. Improved scores on the Revised Neuropsychiatric Inventory (NPI-X) will indicate improvement in these areas.”

What is measured, and how it is judged, sits inside the sentence.



Data

Choices from a list, numbers, links. A system can check and reuse them.

Data in a string

An identifier or a duration written as text. Usable only if you know the convention.

Words

Sentences and written rules. A person has to read them.

Each reading here comes from USDM v4 represented as a knowledge graph.

Five design elements: the schedule is data, the rest is mostly words

Data · Data in a string · Words

Objective ↗

Level · Its endpoints

The objective, written

Endpoint ↗

Level

The endpoint, written · Its purpose

Eligibility criterion ↗

Inclusion or exclusion · Order

Label like “I1”

The criterion, written

no link to its section

Protocol document, its sections ↗

no link in USDM

Study design

Schedule of activities ↗

	Visit	Visit	Visit	Visit
Activity	●	●		●
Activity		●	●	●
Activity	●			●

visit type, contact mode

what is measured

timing and windows

Conditions for branching

Visit start and end rules

Arms × epochs ↗

	Screening	Treatment	Follow-up
Arm 1	element	element	element
Arm 2	element	element	element

Inside each element: what happens, start rule, end rule

In short

USDM gives every design element an identity. Much of the content is still words.

Identity, for the objective on slide 3

USDM, the kind	name `Objective` · C-code `C142450` (Study Objective)
USDM, this one	id `Objective_3` · name `OBJ3` · unique within the study file
usdm-rdf adds	a global identifier for the kind, linked to NCIt C142450: `https://w3id.org/cdisc/usdm/v4/Objective`

Held as data	The schedule of activities, and the arm and epoch grid
Data in a string	Timing values and windows; labels like “I1”
Held as words	Objectives, endpoints, eligibility criteria, rules inside elements
Not held	Endpoint to assessments; design element to protocol section

Beyond USDM

The words are not a dead end

A knowledge graph can represent the entities and relationships from study design to data capture. Structuring what the words say in such a graph turns them into data, and that is where the link from an endpoint to its assessments can live.

